CALCULUS II, QUIZ-2, 16/10/25 Name: Arjun Maneesh Agarwal

Evaluate the following integrals.

1. $\int_A \sin(y^2)$, where A is the region in the plane bounded by the curves $x+y=0, 2x-y=0$ and $y=4$.

2. $\int_0^1 \int_0^{\sqrt{x}} ye^{\sqrt{x}}\,dy\,dx$.

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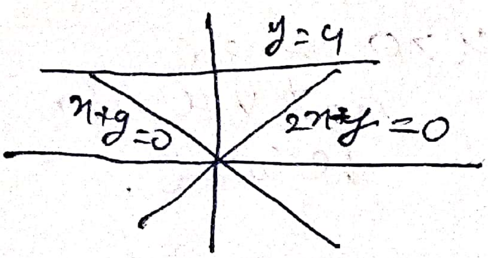
1.



$\Rightarrow$ $0 \le y \le 4$

$-y \le x \le \frac{y}{2}$

$\therefore \int_A \sin(y^2) = \int_{y=0}^{4} \int_{x=-y}^{\frac{y}{2}} \sin(y^2)\,dx\,dy$

$= \int_{y=0}^{4} \sin(y^2)\,x \Big]_{x=-y}^{y/2} dy$

$= \int_{y=0}^{4} \sin(y^2)\,\frac{3y}{2}\,dy$

let $y^2 = u$

$\cancel{2dy = du}$

$\Rightarrow 2y\,dy = du$

$\therefore$ $y=0 \Rightarrow u=0$

$y=4 \Rightarrow u=16$

$= \int_{u=0}^{16} \sin(u)\,\frac{3}{2}\,\frac{du}{2}$

$= \frac{3}{4}\left(-\cos(u)\right)\Big]_0^{16} = \boxed{\frac{3}{4}\left(1-\cos(16)\right)}$